# Variance of the Difference-in-Means Estimator for the PATE and Estimation of That Variance*

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This guide extends the finite sample results to a superpopulation. The guide derives the variance of the Difference-in-Means estimator for the Population Average Treatment Effect (PATE) and then shows that the *same* estimator of the variance that is merely *conservative* for the Sample Average Treatment Effect (SATE) is *exactly unbiased* for the PATE. This guide states the framework only as far as its derivations require and shares that framework with three companion guides: "Notation and Setup for Design-Based Causal Inference" develops the notation in full, "Unbiasedness of the Difference-in-Means Estimator for the SATE and the PATE" establishes that the estimator is unbiased, and "Variance of the Difference-in-Means Estimator for the SATE and Estimation of That Variance" derives the finite sample variance. This guide recalls the key results of those guides where its derivations need them rather than reproving them.

## 1 Setup

Thus far (in the companion SATE guides) I have considered estimation in only a finite sample. Now consider a superpopulation, $\mathcal{P}_N$, of size $N \geq n \geq 4$, and let the index $i \in \{1, \ldots, N\}$ run over the $N$ units in $\mathcal{P}_N$. Let simple random sampling select $n$ units from $\mathcal{P}_N$ into an experimental sample, $\mathcal{S}_n$, leaving the remaining $N - n$ units unsampled. Of these $n$ sampled units, complete random assignment sends $n_T \geq 2$ to treatment and $n - n_T = n_C \geq 2$ to control; the setup guide explains why each arm needs at least two units (each arm's sample variance in the estimator of the variance must be well defined). As in the finite sample setting, each unit $i$ has two *fixed* potential

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outcomes, $y_i(1)$ and $y_i(0)$, with individual effect $\tau_i \coloneqq y_i(1) - y_i(0)$, and the observable outcome is $Y_i \coloneqq y_i(Z_i) = Z_i y_i(1) + (1 - Z_i) y_i(0)$.

The binary indicator variable $R_i \in \{0, 1\}$ records whether unit $i$ enters the random sample from the superpopulation $\mathcal{P}_N$ ($R_i = 1$) or stays out of it ($R_i = 0$). Write $n(\boldsymbol{r}) \coloneqq \sum_{i=1}^{N} r_i$ for the number of units a sampling vector selects. The support of $\boldsymbol{R} = \begin{bmatrix} R_1, \ldots, R_N \end{bmatrix}^\top$ is then $\Pi = \{\boldsymbol{r} \in \{0,1\}^N : n(\boldsymbol{r}) = n\}$, which holds the count of sampled units fixed exactly as $\Omega$ holds the count of treated units fixed: Simple random sampling fixes $n(\boldsymbol{R}) = n$ with probability one, just as complete random assignment fixes $n_T(\boldsymbol{Z}) = n_T$, so both counts appear below as the plain constants $n$ and $n_T$. The setup guide records what licenses a single $\Omega$ across samples — a fixed sample size and a sample-invariant assignment design — and the two-stage stability assumption under which every unit of $\mathcal{P}_N$ has fixed potential outcomes whether or not the sampling stage selects it.

The target of interest is the Population Average Treatment Effect (PATE), $\tau_{\text{PATE}} \coloneqq N^{-1} \sum_{i=1}^{N} \tau_i$. Define the Difference-in-Means estimator of $\tau_{\text{PATE}}$ as

$$\hat{\tau} \coloneqq \frac{1}{n_T} \sum_{i=1}^{N} R_i Z_i Y_i - \frac{1}{n_C} \sum_{i=1}^{N} R_i \left(1 - Z_i\right) Y_i. \tag{1}$$

Here the product $R_i Z_i$ equals 1 only for units that are both sampled and assigned to treatment, and $R_i(1 - Z_i)$ equals 1 only for units that are both sampled and assigned to control; unsampled units have $R_i = 0$ and so contribute nothing, regardless of $Z_i$. Otherwise the setup is identical to the finite sample setup, except that now, under simple random sampling of $n$ units from a population of size $N$ and complete random assignment with $n_T$ treated units out of $n$ sampled units, there are $\binom{N}{n}\binom{n}{n_T}$ possible joint configurations: the $\binom{n}{n_T}$ assignments for any single realized sample, times the $\binom{N}{n}$ samples that could occur.

The estimator of the PATE in (1) has *two* sources of randomness, $\{R_i\}_{i=1}^{N}$ and $\{Z_i\}_{i=1}^{n}$, reflecting the random sampling process and the random assignment process. By contrast, the same estimator in the SATE guides has only the single source $\{Z_i\}_{i=1}^{n}$. For the overall expectation and variance of $\hat{\tau}$ I write $\mathrm{E}[\cdot]$ and $\mathrm{Var}[\cdot]$; I write $\mathrm{E}_\Omega[\cdot], \mathrm{Var}_\Omega[\cdot]$ for expectation and variance over the assignment process (conditional on the sample), and $\mathrm{E}_\Pi[\cdot], \mathrm{Var}_\Pi[\cdot]$ for those over the sampling process. By the law of total variance, the variance of $\hat{\tau}$ about the PATE is

$$\mathrm{Var}\left[\hat{\tau}\right] = \mathrm{E}_\Pi\left[\mathrm{Var}_\Omega\left[\hat{\tau}\right]\right] + \mathrm{Var}_\Pi\left[\mathrm{E}_\Omega\left[\hat{\tau}\right]\right]. \tag{2}$$

I will also need two ingredients from the finite sample (SATE) analysis. First, conditional on the realized sample, the finite sample variances of the potential outcomes (computed over the $n$ sampled units, with divisor $n-1$) are

$$S_n^2(\boldsymbol{y}(1)) = \left(\frac{1}{n-1}\right)\sum_{i=1}^{n}\left(y_i(1) - \frac{1}{n}\sum_{i=1}^{n} y_i(1)\right)^2 \tag{3}$$

$$S_n^2(\boldsymbol{y}(0)) = \left(\frac{1}{n-1}\right)\sum_{i=1}^{n}\left(y_i(0) - \frac{1}{n}\sum_{i=1}^{n} y_i(0)\right)^2 \tag{4}$$

$$S_n^2(\boldsymbol{\tau}) = \left(\frac{1}{n-1}\right)\sum_{i=1}^{n}\left(\tau_i - \frac{1}{n}\sum_{i=1}^{n} \tau_i\right)^2. \tag{5}$$

Second, as the companion SATE variance guide derives, conditional on the realized sample the assignment variance of $\hat{\tau}$ is

$$\mathrm{Var}_\Omega\left[\hat{\tau}\right] = \frac{S_n^2(\boldsymbol{y}(1))}{n_T} + \frac{S_n^2(\boldsymbol{y}(0))}{n_C} - \frac{S_n^2(\boldsymbol{\tau})}{n}, \tag{6}$$

and $\hat{\tau}$ is unbiased for the (sample) SATE, $\mathrm{E}_\Omega[\hat{\tau}] = \tau_{\mathrm{SATE}}$; the proof of Proposition 1 makes precise how that fixed quantity becomes random once the sample itself is random.

## 2 Variance of the Difference-in-Means estimator for the PATE

**Proposition 1.** *The variance of $\hat{\tau}$ for $\tau_{\mathrm{PATE}}$ under simple random sampling from the units in $\mathcal{P}_N$ and complete random assignment among the units in $\mathcal{S}_n$ is*

$$\mathrm{Var}\left[\hat{\tau}\right] = \frac{S_N^2(\boldsymbol{y}(1))}{n_T} + \frac{S_N^2(\boldsymbol{y}(0))}{n_C} - \frac{S_N^2(\boldsymbol{\tau})}{N}, \tag{7}$$

*where the population variances use the divisor $N-1$, in keeping with the naming rule of the setup guide:*

$$S_N^2(\boldsymbol{y}(1)) = \left(\frac{1}{N-1}\right)\sum_{i=1}^{N}\left(y_i(1) - \frac{1}{N}\sum_{i=1}^{N} y_i(1)\right)^2$$

$$S_N^2(\boldsymbol{y}(0)) = \left(\frac{1}{N-1}\right)\sum_{i=1}^{N}\left(y_i(0) - \frac{1}{N}\sum_{i=1}^{N} y_i(0)\right)^2$$

$$S_N^2(\boldsymbol{\tau}) = \left(\frac{1}{N-1}\right)\sum_{i=1}^{N}\left(y_i(1) - y_i(0) - \frac{1}{N}\sum_{i=1}^{N} \tau_i\right)^2.$$

*Proof.* By the law of total variance, Equation (2), the variance of the Difference-in-Means estimator of the PATE is

$$\operatorname{Var}[\hat{\tau}] = \operatorname{E}_{\Pi}\left[\operatorname{Var}_{\Omega}[\hat{\tau}]\right] + \operatorname{Var}_{\Pi}\left[\operatorname{E}_{\Omega}[\hat{\tau}]\right].$$

Since $\hat{\tau}$ is unbiased for the SATE over $\Omega$, $\operatorname{E}_{\Omega}[\hat{\tau}] = \tau_{\text{SATE}}(\boldsymbol{R})$. Here the notation must register a shift in what is random, and the shift deserves its own numbered steps.

**Step 1 (finite population object).** In the SATE guides, $\tau_{\text{SATE}} \coloneqq n^{-1}\sum_{i\in\mathcal{S}_n}\tau_i$ is a fixed number: The sample is given, and nothing about the sample is random.

**Step 2 (new randomness enters).** In this guide the sampling stage determines which units compose $\mathcal{S}_n$, so the same average becomes a function of the sampling vector: $\tau_{\text{SATE}}(\boldsymbol{R}) \coloneqq \frac{1}{n}\sum_{i=1}^{N} R_i\tau_i$ is a random variable, and $\operatorname{Var}_{\Pi}[\tau_{\text{SATE}}(\boldsymbol{R})]$ is a meaningful, nonzero quantity.

**Step 3 (conditioning recovers the fixed object).** Given $\boldsymbol{R} = \boldsymbol{r}$, the value $\tau_{\text{SATE}}(\boldsymbol{r})$ is again a fixed number, and the finite population analysis applies verbatim within the realized sample.

Having stated that shift once, I display the argument wherever the randomness is the point, namely inside $\operatorname{E}_{\Pi}[\cdot]$ and $\operatorname{Var}_{\Pi}[\cdot]$. The variability of $\tau_{\text{SATE}}(\boldsymbol{R})$ across samples is exactly the second component of the law of total variance, $\operatorname{Var}_{\Pi}[\operatorname{E}_{\Omega}[\hat{\tau}]] = \operatorname{Var}_{\Pi}[\tau_{\text{SATE}}(\boldsymbol{R})]$. Using the assignment variance recalled in (6), it follows that

$$\begin{aligned}
\operatorname{Var}[\hat{\tau}] &= \operatorname{E}_{\Pi}\left[\operatorname{Var}_{\Omega}[\hat{\tau}]\right] + \operatorname{Var}_{\Pi}\left[\operatorname{E}_{\Omega}[\hat{\tau}]\right] \\
&= \operatorname{E}_{\Pi}\left[\frac{S_n^2(\boldsymbol{y}(1))}{n_T} + \frac{S_n^2(\boldsymbol{y}(0))}{n_C} - \frac{S_n^2(\boldsymbol{\tau})}{n}\right] + \operatorname{Var}_{\Pi}\left[\tau_{\text{SATE}}(\boldsymbol{R})\right].
\end{aligned}$$

Therefore, I need to derive $\operatorname{E}_{\Pi}[\frac{S_n^2(\boldsymbol{y}(1))}{n_T} + \frac{S_n^2(\boldsymbol{y}(0))}{n_C} - \frac{S_n^2(\boldsymbol{\tau})}{n}]$ and $\operatorname{Var}_{\Pi}[\tau_{\text{SATE}}(\boldsymbol{R})]$.

Elementary theory from survey sampling (Cochran, 1977; Kish, 1965; Lohr, 2010) implies that

$$\operatorname{Var}_{\Pi}\left[\tau_{\text{SATE}}(\boldsymbol{R})\right] = \operatorname{Var}_{\Pi}\left[\frac{1}{n}\sum_{i=1}^{N} R_i\tau_i\right] = \frac{N-n}{N}\,\frac{S_N^2(\boldsymbol{\tau})}{n}.$$

Now I only need to derive $\operatorname{E}_{\Pi}[\frac{S_n^2(\boldsymbol{y}(1))}{n_T} + \frac{S_n^2(\boldsymbol{y}(0))}{n_C} - \frac{S_n^2(\boldsymbol{\tau})}{n}]$:

$$\operatorname{E}_{\Pi}\left[\frac{S_n^2(\boldsymbol{y}(1))}{n_T} + \frac{S_n^2(\boldsymbol{y}(0))}{n_C} - \frac{S_n^2(\boldsymbol{\tau})}{n}\right] = \frac{1}{n_T}\operatorname{E}_{\Pi}\left[S_n^2(\boldsymbol{y}(1))\right] + \frac{1}{n_C}\operatorname{E}_{\Pi}\left[S_n^2(\boldsymbol{y}(0))\right] - \frac{1}{n}\operatorname{E}_{\Pi}\left[S_n^2(\boldsymbol{\tau})\right],$$

since simple random sampling fixes the count $n(\boldsymbol{R})$ at $n$ and complete random assignment fixes the count $n_T(\boldsymbol{Z})$ at $n_T$, so that $n$, $n_T$ and $n_C$ are design constants and pass outside $\operatorname{E}_{\Pi}[\cdot]$. Recalling the definitions of $S_n^2(\boldsymbol{y}(1))$, $S_n^2(\boldsymbol{y}(0))$ and $S_n^2(\boldsymbol{\tau})$ in (3), (4) and (5), we can rewrite each as a function of the sample inclusion indicators $\{R_i\}_{i=1}^N$:

$$S_n^2(\boldsymbol{y}(1)) = \left(\frac{1}{n-1}\right) \sum_{i=1}^N R_i \left(y_i(1) - \frac{1}{n}\sum_{i=1}^N R_i y_i(1)\right)^2$$
$$S_n^2(\boldsymbol{y}(0)) = \left(\frac{1}{n-1}\right) \sum_{i=1}^N R_i \left(y_i(0) - \frac{1}{n}\sum_{i=1}^N R_i y_i(0)\right)^2$$
$$S_n^2(\boldsymbol{\tau}) = \left(\frac{1}{n-1}\right) \sum_{i=1}^N R_i \left(\tau_i - \frac{1}{n}\sum_{i=1}^N R_i \tau_i\right)^2,$$

which make explicit that the randomness of these three sample variances (all of which would be fixed in a finite sample setting) stems from the $N$ sample inclusion indicators $\{R_i\}_{i=1}^N$. They remain *sample* variances, computed over the $n$ sampled units; I keep the subscript $n$ to distinguish them from the population variances $S_N^2(\cdot)$, which are computed over all $N$ units in $\mathcal{P}_N$.

By Cochran (1977, Theorem 2.4), under simple random sampling the sample variance with divisor $n-1$ is *exactly* unbiased for the population variance with divisor $N-1$; matching divisors are precisely what make the finite population correction factor disappear. Hence

$$\mathrm{E}_\Pi\left[S_n^2(\boldsymbol{y}(1))\right] = S_N^2(\boldsymbol{y}(1))$$
$$\mathrm{E}_\Pi\left[S_n^2(\boldsymbol{y}(0))\right] = S_N^2(\boldsymbol{y}(0))$$
$$\mathrm{E}_\Pi\left[S_n^2(\boldsymbol{\tau})\right] = S_N^2(\boldsymbol{\tau}),$$

which implies that

$$\mathrm{E}_\Pi\left[\frac{S_n^2(\boldsymbol{y}(1))}{n_T} + \frac{S_n^2(\boldsymbol{y}(0))}{n_C} - \frac{S_n^2(\boldsymbol{\tau})}{n}\right] = \frac{S_N^2(\boldsymbol{y}(1))}{n_T} + \frac{S_N^2(\boldsymbol{y}(0))}{n_C} - \frac{S_N^2(\boldsymbol{\tau})}{n}.$$

With these two expressions in hand, it follows that

$$\begin{aligned}
\mathrm{Var}\left[\hat{\tau}\right] &= \mathrm{E}_\Pi\left[\mathrm{Var}_\Omega\left[\hat{\tau}\right]\right] + \mathrm{Var}_\Pi\left[\mathrm{E}_\Omega\left[\hat{\tau}\right]\right] \\
&= \frac{S_N^2(\boldsymbol{y}(1))}{n_T} + \frac{S_N^2(\boldsymbol{y}(0))}{n_C} - \frac{S_N^2(\boldsymbol{\tau})}{n} + \frac{N-n}{N}\frac{S_N^2(\boldsymbol{\tau})}{n} \\
&= \frac{S_N^2(\boldsymbol{y}(1))}{n_T} + \frac{S_N^2(\boldsymbol{y}(0))}{n_C} - S_N^2(\boldsymbol{\tau})\left(\frac{1}{n} - \frac{N-n}{Nn}\right) \\
&= \frac{S_N^2(\boldsymbol{y}(1))}{n_T} + \frac{S_N^2(\boldsymbol{y}(0))}{n_C} - \frac{S_N^2(\boldsymbol{\tau})}{N},
\end{aligned}$$

where the last equality uses $\frac{1}{n} - \frac{N-n}{Nn} = \frac{N-(N-n)}{Nn} = \frac{n}{Nn} = \frac{1}{N}$. $\square$

Equation (7) shows that as $N \to \infty$ (with the population variances bounded), the term $\frac{S_N^2(\boldsymbol{\tau})}{N} \to 0$. Write $\sigma^2(\boldsymbol{y}(1))$, $\sigma^2(\boldsymbol{y}(0))$ and $\sigma^2(\boldsymbol{\tau})$ for the superpopulation variances of the treatment potential outcomes, the control potential outcomes, and the individual effects (the limiting values of $S_N^2(\cdot)$ as the superpopulation grows, equivalently the variances of a single unit's quantities under independent sampling from the superpopulation). In that limit the variance of $\hat{\tau}$ about the PATE becomes

$$\text{Var}\left[\hat{\tau}\right] = \frac{\sigma^2(\boldsymbol{y}(1))}{n_T} + \frac{\sigma^2(\boldsymbol{y}(0))}{n_C}. \tag{8}$$

Equation (8) is the variance of the Difference-in-Means estimator about the PATE under the standard superpopulation (or "infinite population") sampling model, which regards the $n$ study units as independent draws from the superpopulation. Unlike the finite sample (SATE) variance, Equation (8) has *no* correction term for effect heterogeneity. The additional sampling variability of $\hat{\tau}$ about the PATE absorbs the $-\,\sigma^2(\boldsymbol{\tau})/\cdot$ term that the finite population expression contains. Because that correction term disappears, the conservative estimator is not merely conservative for the PATE but *exactly unbiased*, as I now show.

## 3 Estimating the variance for the PATE: The conservative estimator is exactly unbiased

In the finite sample (SATE) analysis, the *same* estimator I am about to use was *conservative*, overstating the variance of $\hat{\tau}$ about the SATE by exactly $S_n^2(\boldsymbol{\tau})/n$. I now establish the complementary fact that the same estimator is *exactly unbiased*, not merely conservative, for the variance of $\hat{\tau}$ about the PATE. I need no new estimator. Recall the estimator from the SATE guide,

$$\widehat{\text{Var}}\left[\hat{\tau}\right] = \frac{1}{n_T}\widehat{S}_n^2(\boldsymbol{y}(1)) + \frac{1}{n_C}\widehat{S}_n^2(\boldsymbol{y}(0)), \tag{9}$$

where the sample variances within each group, computed from the sampled units, are

$$\widehat{S}_n^2(\boldsymbol{y}(1)) = \left(\frac{1}{n_T - 1}\right)\sum_{i=1}^{N} R_i Z_i \left(Y_i - \frac{1}{n_T}\sum_{i=1}^{N} R_i Z_i Y_i\right)^2$$
$$\widehat{S}_n^2(\boldsymbol{y}(0)) = \left(\frac{1}{n_C - 1}\right)\sum_{i=1}^{N} R_i (1 - Z_i) \left(Y_i - \frac{1}{n_C}\sum_{i=1}^{N} R_i (1 - Z_i) Y_i\right)^2.$$

**Proposition 2.** *Suppose independent random sampling from the superpopulation produces the $n$ study units (equivalently, take $N \to \infty$ in the finite population setup above), and suppose complete random assignment assigns treatment. Then the estimator $\widehat{\text{Var}}[\hat{\tau}]$ in (9) is exactly unbiased for the* *variance of $\hat{\tau}$ about the PATE:*

$$\mathrm{E}\left[\widehat{\mathrm{Var}}\left[\hat{\tau}\right]\right] = \frac{\sigma^2(\boldsymbol{y}(1))}{n_T} + \frac{\sigma^2(\boldsymbol{y}(0))}{n_C} = \mathrm{Var}\left[\hat{\tau}\right].$$

*That is, the same estimator that is conservative for the SATE is exactly right for the PATE.*

*Proof.* I compute $\mathrm{E}[\widehat{\mathrm{Var}}[\hat{\tau}]]$ by iterating expectations over the assignment process (conditional on the realized sample) and then the sampling process, $\mathrm{E}[\widehat{\mathrm{Var}}[\hat{\tau}]] = \mathrm{E}_\Pi[\mathrm{E}_\Omega[\widehat{\mathrm{Var}}[\hat{\tau}]]]$. Three ingredients do the work.

**Ingredient 1 (the conditional, finite sample result).** Condition on the realized sample. Within it, the analysis is exactly the finite sample (SATE) analysis, and the sample variances within each group are unbiased for the sample variances over the $n$ sampled units (Cochran, 1977, Theorem 2.4), so $\mathrm{E}_\Omega[\widehat{S}_n^2(\boldsymbol{y}(1))] = S_n^2(\boldsymbol{y}(1))$ and $\mathrm{E}_\Omega[\widehat{S}_n^2(\boldsymbol{y}(0))] = S_n^2(\boldsymbol{y}(0))$. Hence the estimator is *conditionally* conservative:

$$\mathrm{E}_\Omega\left[\widehat{\mathrm{Var}}\left[\hat{\tau}\right]\right] = \frac{S_n^2(\boldsymbol{y}(1))}{n_T} + \frac{S_n^2(\boldsymbol{y}(0))}{n_C} = \mathrm{Var}_\Omega\left[\hat{\tau}\right] + \frac{S_n^2(\boldsymbol{\tau})}{n},$$

where the second equality is the variance of $\hat{\tau}$ about the SATE, (6). The conditional overstatement is exactly $S_n^2(\boldsymbol{\tau})/n$.

**Ingredient 2 (law of total variance).** The variance of $\hat{\tau}$ about the PATE decomposes as

$$\mathrm{Var}\left[\hat{\tau}\right] = \mathrm{E}_\Pi\left[\mathrm{Var}_\Omega\left[\hat{\tau}\right]\right] + \mathrm{Var}_\Pi\left[\mathrm{E}_\Omega\left[\hat{\tau}\right]\right] = \mathrm{E}_\Pi\left[\mathrm{Var}_\Omega\left[\hat{\tau}\right]\right] + \mathrm{Var}_\Pi\left[\tau_{\mathrm{SATE}}(\boldsymbol{R})\right],$$

using $\mathrm{E}_\Omega[\hat{\tau}] = \tau_{\mathrm{SATE}}(\boldsymbol{R})$.

**Ingredient 3 (i.i.d. sampling moments).** Here i.i.d. abbreviates *independent and identically distributed*: Each study unit enters the sample independently of the others, and every unit comes from the same superpopulation distribution. Because the $n$ study units are independent draws from the superpopulation, the SATE $\tau_{\mathrm{SATE}}(\boldsymbol{R})$ is the mean of $n$ independent draws of the individual effect, and $S_n^2(\boldsymbol{\tau})$ is the corresponding sample variance. Hence

$$\mathrm{Var}_\Pi\left[\tau_{\mathrm{SATE}}(\boldsymbol{R})\right] = \frac{\sigma^2(\boldsymbol{\tau})}{n} \quad \text{and} \quad \mathrm{E}_\Pi\left[S_n^2(\boldsymbol{\tau})\right] = \sigma^2(\boldsymbol{\tau}),$$

both of which are standard facts for an i.i.d. sample, the first for the variance of a sample mean and the second for the unbiasedness of the sample variance. (The same reasoning gives $\mathrm{E}_\Pi[S_n^2(\boldsymbol{y}(1))] = \sigma^2(\boldsymbol{y}(1))$ and $\mathrm{E}_\Pi[S_n^2(\boldsymbol{y}(0))] = \sigma^2(\boldsymbol{y}(0))$.)

Now take the expectation of Ingredient 1 over the sampling distribution:

$$\begin{aligned}
\mathrm{E}\left[\widehat{\mathrm{Var}}\left[\hat{\tau}\right]\right] &= \mathrm{E}_{\Pi}\left[\mathrm{E}_{\Omega}\left[\widehat{\mathrm{Var}}\left[\hat{\tau}\right]\right]\right] = \mathrm{E}_{\Pi}\left[\mathrm{Var}_{\Omega}\left[\hat{\tau}\right] + \frac{S_n^2(\boldsymbol{\tau})}{n}\right] \\
&= \mathrm{E}_{\Pi}\left[\mathrm{Var}_{\Omega}\left[\hat{\tau}\right]\right] + \frac{\mathrm{E}_{\Pi}\left[S_n^2(\boldsymbol{\tau})\right]}{n} = \mathrm{E}_{\Pi}\left[\mathrm{Var}_{\Omega}\left[\hat{\tau}\right]\right] + \frac{\sigma^2(\boldsymbol{\tau})}{n},
\end{aligned}$$

using Ingredient 3 in the last step. Subtracting the variance of $\hat{\tau}$ about the PATE from Ingredient 2,

$$\begin{aligned}
\mathrm{E}\left[\widehat{\mathrm{Var}}\left[\hat{\tau}\right]\right] - \mathrm{Var}\left[\hat{\tau}\right] &= \left(\mathrm{E}_{\Pi}\left[\mathrm{Var}_{\Omega}\left[\hat{\tau}\right]\right] + \frac{\sigma^2(\boldsymbol{\tau})}{n}\right) - \left(\mathrm{E}_{\Pi}\left[\mathrm{Var}_{\Omega}\left[\hat{\tau}\right]\right] + \mathrm{Var}_{\Pi}\left[\tau_{\mathrm{SATE}}(\boldsymbol{R})\right]\right) \\
&= \frac{\sigma^2(\boldsymbol{\tau})}{n} - \mathrm{Var}_{\Pi}\left[\tau_{\mathrm{SATE}}(\boldsymbol{R})\right] \\
&= \frac{\sigma^2(\boldsymbol{\tau})}{n} - \frac{\sigma^2(\boldsymbol{\tau})}{n} = 0,
\end{aligned}$$

again by Ingredient 3. Hence $\mathrm{E}[\widehat{\mathrm{Var}}[\hat{\tau}]] = \mathrm{Var}[\hat{\tau}]$. Computing the common value directly, take expectations of Ingredient 1 and use $\mathrm{E}_{\Pi}[S_n^2(\boldsymbol{y}(1))] = \sigma^2(\boldsymbol{y}(1))$ and $\mathrm{E}_{\Pi}[S_n^2(\boldsymbol{y}(0))] = \sigma^2(\boldsymbol{y}(0))$:

$$\mathrm{E}\left[\widehat{\mathrm{Var}}\left[\hat{\tau}\right]\right] = \mathrm{E}_{\Pi}\left[\frac{S_n^2(\boldsymbol{y}(1))}{n_T} + \frac{S_n^2(\boldsymbol{y}(0))}{n_C}\right] = \frac{\sigma^2(\boldsymbol{y}(1))}{n_T} + \frac{\sigma^2(\boldsymbol{y}(0))}{n_C} = \mathrm{Var}\left[\hat{\tau}\right],$$

the last equality by (8). □

The cancellation in the proof has a simple interpretation. Conditional on the sample, the conservative estimator overshoots the variance of $\hat{\tau}$ about the SATE by exactly $S_n^2(\boldsymbol{\tau})/n$, the penalty for effect heterogeneity that makes the estimator conservative for the SATE. Once the estimand is the PATE, the estimator must also contend with a source of variability that the SATE analysis does not face, namely the randomness in which units are sampled, which makes the sample's SATE fluctuate about the PATE with variance $\mathrm{Var}_{\Pi}[\tau_{\mathrm{SATE}}(\boldsymbol{R})] = \sigma^2(\boldsymbol{\tau})/n$. In expectation the conditional overstatement $S_n^2(\boldsymbol{\tau})/n$ equals the extra sampling variance $\sigma^2(\boldsymbol{\tau})/n$. The slack that makes the estimator conservative for the SATE is precisely the variability the estimator must absorb to be exactly right for the PATE.

In a finite population the exact unbiasedness just established fails slightly. Suppose that, instead of taking independent draws from the superpopulation, simple random sampling *without replacement* draws the $n$ units from a *finite* population of fixed size $N$, the setting of Proposition 1. Then the conditional overstatement and the extra sampling variance no longer cancel exactly. A finite population correction leaves a residual $\mathrm{E}_{\Pi}[S_n^2(\boldsymbol{\tau})]/n - \mathrm{Var}_{\Pi}[\tau_{\mathrm{SATE}}(\boldsymbol{R})] = S_N^2(\boldsymbol{\tau})/n - \frac{N-n}{N} S_N^2(\boldsymbol{\tau})/n = S_N^2(\boldsymbol{\tau})/N \geq 0$, so the estimator is mildly conservative, with bias $S_N^2(\boldsymbol{\tau})/N$. This residual vanishes as $N \to \infty$, which recovers the exact unbiasedness above. The exact result is therefore the one that pertains to the PATE proper, the average effect in the (effectively infinite) superpopulation from which studies draw their samples.

## 4 Extension: simple random assignment within the sample

Throughout, the assignment stage has used complete random assignment, with $n_T$ fixed by design. Suppose instead that simple random assignment assigns treatment within the realized sample — each unit independently, with a common assignment probability (see the setup guide) — so that the count $n_T(\boldsymbol{Z})$ is a nondegenerate random variable and the estimator uses the realized counts,

$$\hat{\tau} = \frac{1}{n_T(\boldsymbol{Z})} \sum_{i=1}^{N} R_i Z_i Y_i - \frac{1}{n_C(\boldsymbol{Z})} \sum_{i=1}^{N} R_i \left(1 - Z_i\right) Y_i.$$

The sampling stage is unchanged: Simple random sampling still draws exactly $n$ units. Because every realized sample has the same size and faces the same assignment mechanism, the expected reciprocals $\mathrm{E}_\Omega[1/n_T(\boldsymbol{Z})]$ and $\mathrm{E}_\Omega[1/n_C(\boldsymbol{Z})]$ do not depend on which sample the sampling stage produces.

**Proposition 3.** *Under simple random sampling from the units in $\mathcal{P}_N$ and simple random assignment among the units in $\mathcal{S}_n$,*

$$\mathrm{Var}\left[\hat{\tau}\right] = S_N^2(\boldsymbol{y}(1))\ \mathrm{E}_\Omega\left[\frac{1}{n_T(\boldsymbol{Z})}\right] + S_N^2(\boldsymbol{y}(0))\ \mathrm{E}_\Omega\left[\frac{1}{n_C(\boldsymbol{Z})}\right] - \frac{S_N^2(\boldsymbol{\tau})}{N}.$$

*Proof.* Every ingredient is already in hand. The law of total variance, Equation (2), again decomposes the variance into $\mathrm{E}_\Pi[\mathrm{Var}_\Omega[\hat{\tau}]] + \mathrm{Var}_\Pi[\mathrm{E}_\Omega[\hat{\tau}]]$. The unbiasedness guide's simple random assignment result, applied within the realized sample, still gives $\mathrm{E}_\Omega[\hat{\tau}] = \tau_{\mathrm{SATE}}(\boldsymbol{R})$, so the second component remains $\mathrm{Var}_\Pi[\tau_{\mathrm{SATE}}(\boldsymbol{R})] = \frac{N-n}{N} \frac{S_N^2(\boldsymbol{\tau})}{n}$. For the first component, the extension result of the companion SATE variance guide, applied within the realized sample, gives

$$\mathrm{Var}_\Omega\left[\hat{\tau}\right] = S_n^2(\boldsymbol{y}(1))\ \mathrm{E}_\Omega\left[\frac{1}{n_T(\boldsymbol{Z})}\right] + S_n^2(\boldsymbol{y}(0))\ \mathrm{E}_\Omega\left[\frac{1}{n_C(\boldsymbol{Z})}\right] - \frac{S_n^2(\boldsymbol{\tau})}{n},$$

with the sample variances computed over the sampled units. Taking the expectation over the sampling stage and using $\mathrm{E}_\Pi[S_n^2(\cdot)] = S_N^2(\cdot)$,

$$\mathrm{E}_\Pi\left[\mathrm{Var}_\Omega\left[\hat{\tau}\right]\right] = S_N^2(\boldsymbol{y}(1))\ \mathrm{E}_\Omega\left[\frac{1}{n_T(\boldsymbol{Z})}\right] + S_N^2(\boldsymbol{y}(0))\ \mathrm{E}_\Omega\left[\frac{1}{n_C(\boldsymbol{Z})}\right] - \frac{S_N^2(\boldsymbol{\tau})}{n}.$$

Adding $\mathrm{Var}_\Pi[\tau_{\mathrm{SATE}}(\boldsymbol{R})]$,

$$\begin{aligned}\mathrm{Var}\left[\hat{\tau}\right] = S_N^2(\boldsymbol{y}(1))\ \mathrm{E}_\Omega\left[\frac{1}{n_T(\boldsymbol{Z})}\right] + S_N^2(\boldsymbol{y}(0))\ \mathrm{E}_\Omega\left[\frac{1}{n_C(\boldsymbol{Z})}\right] \\ - \frac{S_N^2(\boldsymbol{\tau})}{n} + \frac{N-n}{N}\frac{S_N^2(\boldsymbol{\tau})}{n},\end{aligned}$$

and the two $S_N^2(\boldsymbol{\tau})$ terms combine exactly as in the proof of Proposition 1: $-\frac{1}{n} + \frac{N-n}{Nn} = \frac{-N+N-n}{Nn} = -\frac{1}{N}$, which yields the claim. □

Because the variance in Proposition 3 depends on the assignment mechanism only through the same expected reciprocals, the comparison from the companion SATE variance guide applies here as well: Fixing balanced counts cannot increase the variance of $\hat{\tau}$ about the PATE.

The estimation result survives as well, conditional on the event $\mathcal{D} := \{2 \le n_T(\boldsymbol{Z}) \le n-2\}$ that each arm contains at least two units. Note that $\mathcal{D}$ concerns only the assignment stage, and the distribution of the count of treated units is the same for every realized sample, so conditioning on $\mathcal{D}$ says nothing about $\boldsymbol{R}$. Within any realized sample, the extension results of the companion SATE variance guide give

$$\mathrm{E}_\Omega\left[\widehat{\mathrm{Var}}\left[\hat{\tau}\right] \mid \mathcal{D}\right] = \mathrm{Var}_\Omega\left[\hat{\tau} \mid \mathcal{D}\right] + \frac{S_n^2(\boldsymbol{\tau})}{n},$$

an overshoot that does not depend on the realized counts, and $\mathrm{E}_\Omega[\hat{\tau} \mid \mathcal{D}] = \tau_{\mathrm{SATE}}(\boldsymbol{R})$, because the conditional expectation of the estimator equals $\tau_{\mathrm{SATE}}(\boldsymbol{R})$ at every count in $\mathcal{D}$. Taking expectations over the sampling stage and applying the law of total variance conditional on $\mathcal{D}$,

$$\begin{aligned}\mathrm{E}\left[\widehat{\mathrm{Var}}\left[\hat{\tau}\right] \mid \mathcal{D}\right] &= \mathrm{E}_\Pi\left[\mathrm{Var}_\Omega\left[\hat{\tau} \mid \mathcal{D}\right]\right] + \frac{\mathrm{E}_\Pi\left[S_n^2(\boldsymbol{\tau})\right]}{n} \\ \mathrm{Var}\left[\hat{\tau} \mid \mathcal{D}\right] &= \mathrm{E}_\Pi\left[\mathrm{Var}_\Omega\left[\hat{\tau} \mid \mathcal{D}\right]\right] + \mathrm{Var}_\Pi\left[\tau_{\mathrm{SATE}}(\boldsymbol{R})\right].\end{aligned}$$

Subtracting the second line from the first leaves $\mathrm{E}_\Pi[S_n^2(\boldsymbol{\tau})]/n - \mathrm{Var}_\Pi[\tau_{\mathrm{SATE}}(\boldsymbol{R})]$, which is exactly the difference that Ingredient 3 of the proof of Proposition 2 evaluates: zero in the infinite superpopulation limit. The same estimator, now computed with the realized counts, remains exactly unbiased for the variance of $\hat{\tau}$ about the PATE.

## References


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